



Lakshya JEE 2027

KPP - 06

Electrostatics

BY: Saleem Bhaiya

1. Two identical conducting spheres P and S with charge Q on each, repel each other with a force 16 N. A third identical uncharged conducting sphere R is successively brought in contact with the two spheres. The new force of repulsion between P and S is:

[JEE MAINS April 6, 2024 (II)]

- (1) 4N (2) 6N
(3) 1N (4) 12N

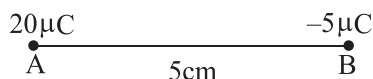
2. Two equal positive point charges are separated by a distance $2a$. The distance of a point from the centre of the line joining two charges on the equatorial line (perpendicular bisector) at which force experienced by a test charge q_0 becomes maximum is $\frac{a}{\sqrt{x}}$. The value of x is _____. [JEE MAINS Feb. 1, 2023 (I)]

3. A charge of $4 \mu\text{C}$ is to be divided into two parts. The distance between the two divided charges is constant. The magnitude of the divided charges so that the force between them is maximum, will be:

[JEE MAINS July 27, 2022 (II)]

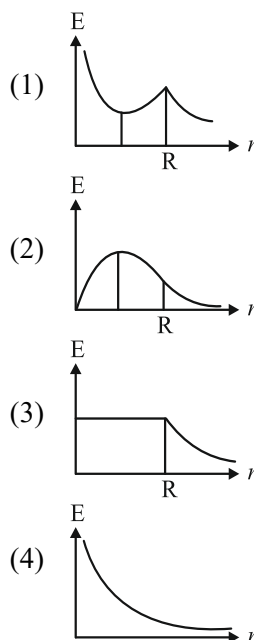
- (1) $1 \mu\text{C}$ and $3 \mu\text{C}$ (2) $2 \mu\text{C}$ and $2 \mu\text{C}$
(3) $0 \mu\text{C}$ and $4 \mu\text{C}$ (4) $1.5 \mu\text{C}$ and $2.5 \mu\text{C}$

4. Two particles A and B having charges $20 \mu\text{C}$ and $-5 \mu\text{C}$ respectively are held fixed with a separation of 5 cm. At what position a third charged particle should be placed so that it does not experience a net electric force? [JEE MAINS Aug. 31, 2021 (I)]

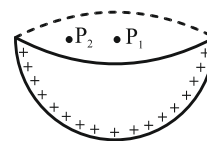


- (1) At 5 cm from $20 \mu\text{C}$ on the left side of system
(2) At 5 cm from $-5 \mu\text{C}$ on the right side
(3) At 1.25 cm from $-5 \mu\text{C}$ between two charges
(4) At midpoint between two charges

5. A spherical insulator of radius R is charged uniformly with a charge Q throughout its volume and contains a point charge $\frac{Q}{16}$ located at its centre. Which of the following graphs best represent qualitatively, the variation of electric field intensity E with distance r from the centre.

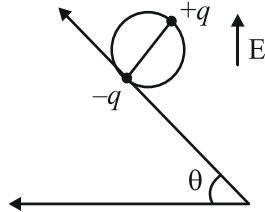


6. Consider a uniformly charged hemispherical shell shown below. Indicate the directions (not magnitude) of the electric field at the central point P_1 , and an off-centre point P_2 on the drumhead of the shell.

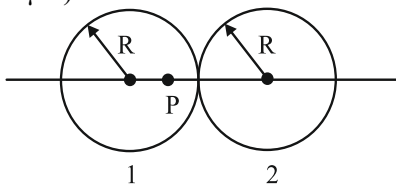


- (1) $\uparrow$; $\nearrow$ (2) $\uparrow$; $\nearrow$
(3) $\uparrow$; $\uparrow$ (4) $\uparrow$; $\leftarrow$

7. A wheel having mass m has charges $+q$ and $-q$ on diametrically opposite points. It remains in equilibrium on a rough inclined plane in the presence of uniform vertical electric field $E =$



- (1) $\frac{mg}{q}$ (2) $\frac{mg}{2q}$
 (3) $\frac{mg \tan \theta}{2q}$ (4) none
8. The surface charge density of a thin charged disc of radius R is σ . The value of the electric field at the centre of the disc is $\frac{\sigma}{2\epsilon_0}$. With respect to the field at the centre, the electric field along the axis at a distance $\sqrt{3}R$ from the centre of the disc:
- (1) reduces by 70.7%
 (2) reduces by 29.3%
 (3) reduces by 86.6%
 (4) reduces by 13.4%
9. Figure shows, in cross section, two solid spheres with uniformly distributed charge throughout their volumes. Each has radius R . Point P lies on a line connecting the centres of the spheres, at radial distance $R/2$ from the center of sphere 1. If the net electric field at point P is zero and Q_1 is $64 \mu\text{C}$, what is Q_2 (in μC).



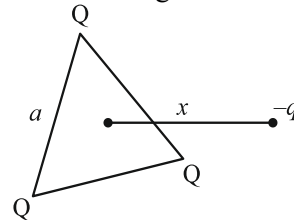
- (1) 64 (2) 36
 (3) 32 (4) 72

10. A sphere of radius R carries charge density proportional to the square of the distance from the center: $\sigma = Ar^2$, where A is a positive constant. At a distance of $R/2$ from the center, the magnitude of the electric field is:

- (1) $A/4\pi\epsilon_0$ (2) $AR^3/40\epsilon_0$
 (3) $AR^3/24\epsilon_0$ (4) $AR^3/5\epsilon_0$

Paragraph for Questions No. 11 to 13

Three charged particles each of $+Q$ are fixed at the corners of an equilateral triangle of side ' a '. A fourth particle of charge $-q$ and mass m is placed at a point on the line passing through centroid of triangle and perpendicular to the plane of triangle at a distance x from the centre of triangle.



11. Magnitude of resultant force on the fourth charged particle is:

- (1) $\frac{1}{4\pi\epsilon_0} \frac{9\sqrt{3} Qqx}{(3x^2 + a^2)^{3/2}}$
 (2) $\frac{1}{4\pi\epsilon_0} \frac{27\sqrt{3} Qqx}{(3x^2 + a^2)^{3/2}}$
 (3) $\frac{1}{4\pi\epsilon_0} \frac{2\sqrt{2} Qqx}{(2x^2 + a^2)^{3/2}}$
 (4) $\frac{1}{4\pi\epsilon_0} \frac{4\sqrt{2} Qqx}{(2x^2 + a^2)^{3/2}}$

12. Value of x for which the force is maximum is:

- (1) $\frac{a}{\sqrt{3}}$ (2) $\frac{a}{\sqrt{2}}$
 (3) $\frac{a}{\sqrt{6}}$ (4) $\frac{a}{\sqrt{5}}$

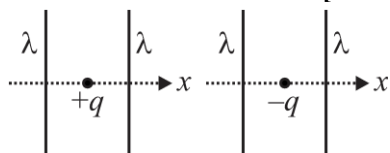
13. For small oscillation the period of oscillation of fourth particle is:

- (1) $2\pi\sqrt{\frac{4\pi\epsilon_0 ma^3}{9\sqrt{3} Qq}}$ (2) $\pi\sqrt{\frac{4\pi\epsilon_0 ma^3}{9\sqrt{3} Qq}}$
 (3) $2\pi\sqrt{\frac{2\pi\epsilon_0 ma^3}{27\sqrt{3} Qq}}$ (4) $2\pi\sqrt{\frac{\pi\epsilon_0 ma^3}{27\sqrt{3} Qq}}$

14. Two non-conducting spheres of radii R_1 and R_2 and carrying uniform volume charge densities $+\rho$ and $-\rho$, respectively, are placed such that they partially overlap, as shown in the figure. At all points in the overlapping region: [JEE-Advance-2013]

- (1) the electrostatic field is zero
- (2) the electrostatic potential is constant
- (3) the electrostatic field is constant in magnitude
- (4) the electrostatic field has same direction

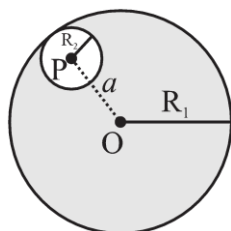
15. The figures below depict two situations in which two infinitely long static line charges of constant positive line charge density λ are kept parallel to each other. In their resulting electric field, point charges q and $-q$ are kept in equilibrium between them. The point charges are confined to move in the x direction only. If they are given a small displacement about their equilibrium positions, then the correct statement(s) is(are) [JEE Adv. - 2015]



- (1) Both charges execute simple harmonic motion
- (2) Both charges will continue moving in the direction of their displacement
- (3) Charge $+q$ executes simple harmonic motion while charge $-q$ continues moving in the direction of its displacement
- (4) Charge $-q$ executes simple harmonic motion while charge $+q$ continues moving in the direction of its displacement

16. Consider a uniform spherical charge distribution of radius R_1 centred at the origin O . In this distribution, a spherical cavity of radius R_2 , centred at P with distance $OP = a = R_1 - R_2$ (see figure) is made. If the electric field inside the cavity at position $\vec{r}$ is $\vec{E}(\vec{r})$, then the correct statement(s) is (are):

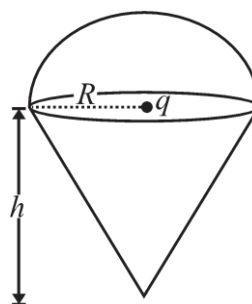
[JEE Adv. - 2015]



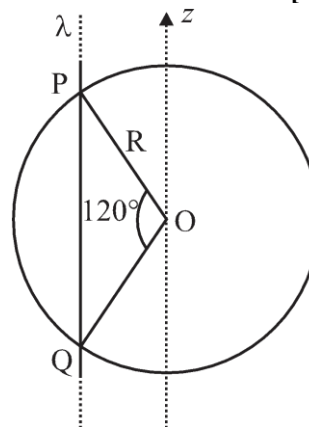
- (1) $\vec{E}$ is uniform, its magnitude is independent of R_2 but its direction depends on $\vec{r}$
- (2) $\vec{E}$ is uniform, its magnitude depends on R_2 and its direction depends on $\vec{r}$
- (3) $\vec{E}$ is uniform, its magnitude is independent of a but its direction depends on $\vec{a}$
- (4) $\vec{E}$ is uniform and both its magnitude and direction depends on $\vec{a}$

17. A charge q is surrounded by a closed surface consisting of an inverted cone of height h and base radius R , and a hemisphere of radius R as shown in the figure. The electric flux through the conical surface is $\frac{nq}{6\epsilon_0}$ (in SI units). The value of n is

[JEE Adv. - 2022]



18. An infinitely long thin non-conducting wire is parallel to the z -axis and carries a uniform line charge density λ . It pierces a thin non-conducting spherical shell of radius R in such a way that the arc PQ subtends an angle 120° at the centre O of the spherical shell, as shown in the figure. The permittivity of free space is ϵ_0 . Which of the following statements is (are) true? [JEE Adv. - 2018]

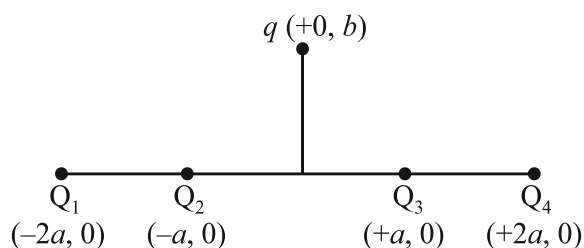




- (1) The electric flux through the shell is $\frac{\sqrt{3}R\lambda}{\epsilon_0}$
- (2) The z -component of the electric field is zero at all the points on the surface of the shell
- (3) The electric flux through the shell is $\frac{\sqrt{2}R\lambda}{\epsilon_0}$
- (4) The electric field is normal to the surface of the shell at all points

19. Four charges Q_1, Q_2, Q_3 and Q_4 of same magnitude are fixed along the $x = -2a, -a, +a$ and $+2a$, respectively. A positive charge q is placed on the positive y -axis at a distance $b > 0$. Four options of the signs of these charges are given in List-I. The direction of the forces on the charge q is given in List-II. Match List-I with List-II and select the correct answer using the code given below the lists.

[JEE Adv. - 2014]



List I		List II	
P	Q_1, Q_2, Q_3, Q_4 all positive	1	$+x$
Q	Q_1, Q_2 positive Q_3, Q_4 negative	2	$-x$
R	Q_1, Q_4 positive Q_2, Q_3 negative	3	$+y$
S	Q_1, Q_3 positive Q_2, Q_4 negative	4	$-y$

- (1) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$
- (2) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$
- (3) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$
- (4) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$



Answer Key

- | | |
|---------|------------|
| 1. (2) | 11. (1) |
| 2. (2) | 12. (3) |
| 3. (2) | 13. (1) |
| 4. (2) | 14. (3, 4) |
| 5. (1) | 15. (3) |
| 6. (3) | 16. (4) |
| 7. (2) | 17. (3) |
| 8. (3) | 18. (1, 2) |
| 9. (4) | 19. (1) |
| 10. (2) | |



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>